The role of animal models in the study of varicocele.

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Varicocele is the most common correctible cause of male infertility and is present in 15% to 20% of the male population. Despite its prevalence, the pathophysiology of varicocele remains under investigation. One of the largest obstacles in studying varicocele is that it is almost exclusively found in humans. This has necessitated the creation of an animal model of varicocele. The most commonly used animal model involves the creation of a varicocele in a rodent by partially occluding the left renal vein. This model has provided a significant amount of data on varicocele, and a modification of this model utilizing microsurgery appears even more promising. Animal models have proven critical to investigating the pathophysiology of varicocele.

Proteomic profile of seminal plasma in adolescents and adults with treated and untreated varicocele.

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Varicocele, the most important treatable cause of male infertility, is present in 15% of adult males, 35% of men with primary infertility, and 80% of men with secondary infertility. On the other hand, 80% of these men will not present infertility. Therefore, there is a need to differentiate a varicocele that is exerting a deleterious effect that is treatable from a "silent" varicocele. Despite the growing evidence of the cellular effects of varicocele, its underlying molecular mechanisms are still eluding. Proteomics has become a promising area to determine the reproductive biology of semen as well as to improve diagnosis of male infertility. This review aims to discuss the state-of-art in seminal plasma proteomics in patients with varicocele to discuss the challenges in undertaking these studies, as well as the future outlook derived from the growing body of evidence on the seminal proteome.

Molecular mechanisms involved in varicocele-associated infertility.

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Varicocele is a pathologic enlargement of the pampiniform venous plexus within the spermatic cord, a condition that is a common cause of impaired sperm production and decreased quality of sperm. While varicocele is the most common surgically correctable risk factor for male infertility, not all males with varicocele experience infertility. In fact, most men with varicocele have normal spermatogenesis. Despite its prevalence, the molecular mechanisms of varicocele and its effect on testicular function are yet to be completely understood. We postulate that men with varicocele-associated infertility could have preexisting genetic lesions or defects in molecular mechanisms that make them more susceptible to varicocele-mediated testicular injury affecting spermatogenesis.

[Varicocele: an epidemiological study and indications for treatment].

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The varicocele is still the most common correctable cause of male infertility. The epidemiologic studies revealed a gradually increasing incidence of varicocele in patients 10 to 18 years old, as height at the end of puberty as that of the adult male population. The pediatric varicocele is often associated with smaller and hypotrophic testis which presents also histological and progressive changes. Surgical correction of a varicocele in the adult usually produces poor results. In order to improve the fertility rate in adulthood an early varicocelectomy has been suggested. On the basis of our own studies and after a review of the literature, we conclude that all grade III varicoceles and grade II associated with testicular hypotrophy must be early operated on. This hearing provides a higher fertility rate and can prevent the reduced fertility associated with delayed varicocelectomy.

Reassessment of male-factor infertility, including the varicocele, sperm penetration assay, semen analysis, and in vitro fertilization.

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Evaluation of the male factor in infertility is becoming increasingly important as new diagnostic techniques and therapeutic options become available. Varicoceles are among the most common treatable cause of male infertility. Varicoceles are present in 10% to 20% of all males but are found in as many as 30% to 40% of men who present to an infertility clinic. Of men who have treatment for varicoceles, 50% to 75% will show some improvement in semen quality, and 30% to 40% will initiate a pregnancy. We review some controversial issues related to the diagnosis and treatment of varicoceles. In vitro fertilization, originally developed for the female with irreversible tubal damage, is now being evaluated as a possible therapy for severe male-factor infertility that has failed to respond to routine surgical or medical treatment. In vitro fertilization for the oligospermic male, however, is further complicated by the fact that men with poor sperm production frequently have poorly functioning sperm as well. Consequently, we discuss the value of the sperm penetration assay, with and without enhancement techniques to prospectively evaluate couples entering in vitro fertilization programs. We also discuss the role of strict criteria for determination of sperm morphology and quantitation of leukocytospermia in the evaluation of the infertile male. Finally, evaluation of the predictive value of "failure to fertilize" at in vitro fertilization for future in vitro fertilization success is discussed.

Adolescent varicocoele management controversies.

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Varicocoele is defined as excessive dilatation of the pampiniform venous plexus of the spermatic cord. Varicocoele frequently appears during early puberty and is recognized to be the most common surgically correctable cause of male infertility. However, the actual incidence in adolescents, pathophysiology and the association with male factor infertility all remain somewhat controversial. The most accurate diagnostic technique for identifying young men who will benefit from surgical treatment has yet to be established. Observations of testicular asymmetry and deteriorating semen quality helped establish current guidelines and recommendations for surgical treatment. Further studies, comparing observation with surgical intervention, are needed to refine the current indications for varicocoele repair in the adolescent male.

Plozasiran (ARO-APOC3) for Severe Hypertriglyceridemia: The SHASTA-2 Randomized Clinical Trial.

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Severe hypertriglyceridemia (sHTG) confers increased risk of atherosclerotic cardiovascular disease (ASCVD), nonalcoholic steatohepatitis, and acute pancreatitis. Despite available treatments, persistent ASCVD and acute pancreatitis-associated morbidity from sHTG remains. To determine the tolerability, efficacy, and dose of plozasiran, an APOC3-targeted small interfering-RNA (siRNA) drug, for lowering triglyceride and apolipoprotein C3 (APOC3, regulator of triglyceride metabolism) levels and evaluate its effects on other lipid parameters in patients with sHTG. The Study to Evaluate ARO-APOC3 in Adults With Severe Hypertriglyceridemia (SHASTA-2) was a placebo-controlled, double-blind, dose-ranging, phase 2b randomized clinical trial enrolling adults with sHTG at 74 centers across the US, Europe, New Zealand, Australia, and Canada from May 31, 2021, to August 31, 2023. Eligible patients had fasting triglyceride levels in the range of 500 to 4000 mg/dL (to convert to millimoles per liter, multiply by 0.0113) while receiving stable lipid-lowering treatment. Participants received 2 subcutaneous doses of plozasiran (10, 25, or 50 mg) or matched placebo on day 1 and at week 12 and were followed up through week 48. The primary end point evaluated the placebo-subtracted difference in means of percentage triglyceride change at week 24. Mixed-model repeated measures were used for statistical modeling. Of 229 patients, 226 (mean [SD] age, 55 [11] years; 176 male [78%]) were included in the primary analysis. Baseline mean (SD) triglyceride level was 897 (625) mg/dL and plasma APOC3 level was 32 (16) mg/dL. Plozasiran induced significant dose-dependent placebo-adjusted least squares (LS)-mean reductions in triglyceride levels (primary end point) of -57% (95% CI, -71.9% to -42.1%; $P < .001$), driven by placebo-adjusted reductions in APOC3 of -77% (95% CI, -89.1% to -65.8%; $P < .001$) at week 24 with the highest dose. Among plozasiran-treated patients, 144 of 159 (90.6%) achieved a triglyceride level of less than 500 mg/dL. Plozasiran was associated with dose-dependent increases in low-density lipoprotein cholesterol (LDL-C) level, which was significant in patients receiving the highest dose (placebo-adjusted LS-mean increase 60% (95% CI, 31%-89%; $P < .001$). However, apolipoprotein B (ApoB) levels did not increase, and non-high-density lipoprotein cholesterol (HDL-C) levels decreased significantly at all doses, with a placebo-adjusted change of -20% at the highest dose. There were also significant durable reductions in remnant cholesterol and ApoB48 as well as increases in HDL-C level through week 48. Adverse event rates were similar in plozasiran-treated patients vs placebo. Serious adverse events were mild to moderate, not considered treatment related, and none led to discontinuation or death. In this randomized clinical trial of patients with sHTG, plozasiran decreased triglyceride levels, which fell below the 500 mg/dL threshold of acute pancreatitis risk in most participants. Other triglyceride-related lipoprotein parameters improved. An increase in LDL-C level was observed but with no change in ApoB level and a decrease in non-HDL-C level. The safety profile was generally favorable at all doses. Additional studies will be required to determine whether plozasiran favorably modulates the risk of sHTG-associated complications. ClinicalTrials.gov Identifier: NCT04720534.

Novel agents for treating severe hypertriglyceridemia.

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The PALISADE trial extended the data available for inhibition of apolipoprotein (apo) C3 inhibition for treating severe hypertriglyceridemia.1 75 patients with persistent chylomicronemia were allocated to 2 doses of plozasiran or placebo. Triglycerides were reduced by a net 53%-58%, and a borderline significant 17% reduction was seen in pancreatitis events. These results offer potential treatments for

familial or multifactorial chylomicronemia syndromes.

Plozasiran for Managing Persistent Chylomicronemia and Pancreatitis Risk.

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Persistent chylomicronemia is a genetic recessive disorder that is classically caused by familial chylomicronemia syndrome (FCS), but it also has multifactorial causes. The disorder is associated with the risk of recurrent acute pancreatitis. Plozasiran is a small interfering RNA that reduces hepatic production of apolipoprotein C-III and circulating triglycerides. In a phase 3 trial, we randomly assigned 75 patients with persistent chylomicronemia (with or without a genetic diagnosis) to receive subcutaneous plozasiran (25 mg or 50 mg) or placebo every 3 months for 12 months. The primary end point was the median percent change from baseline in the fasting triglyceride level at 10 months. Key secondary end points were the percent change in the fasting triglyceride level from baseline to the mean of values at 10 months and 12 months, changes in the fasting apolipoprotein C-III level from baseline to 10 months and 12 months, and the incidence of acute pancreatitis. At baseline, the median triglyceride level was 2044 mg per deciliter. At 10 months, the median change from baseline in the fasting triglyceride level (the primary end point) was -80% in the 25-mg plozasiran group, -78% in the 50-mg plozasiran group, and -17% in the placebo group ($P<0.001$). The key secondary end points showed better results in the plozasiran groups than in the placebo group, including the incidence of acute pancreatitis (odds ratio, 0.17; 95% confidence interval, 0.03 to 0.94; $P = 0.03$). The risk of adverse events was similar across groups; the most common adverse events were abdominal pain, nasopharyngitis, headache, and nausea. Severe and serious adverse events were less common with plozasiran than with placebo. Hyperglycemia with plozasiran occurred in some patients with prediabetes or diabetes at baseline. Patients with persistent chylomicronemia who received plozasiran had significantly lower triglyceride levels and a lower incidence of pancreatitis than those who received placebo. (Funded by Arrowhead Pharmaceuticals; PALISADE ClinicalTrials.gov number, NCT05089084.).

Highlights of Cardiovascular Disease Prevention Studies Presented at the 2024 European Society of Cardiology Congress.

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To summarize selected late-breaking science on cardiovascular disease (CVD) prevention presented at the 2024 European Society of Cardiology (ESC) Congress. Key studies from the 2024 ESC Congress highlight advances in (CVD) management. Apolipoprotein A-1 infusions reduced risk in acute myocardial infarction patients with high LDL cholesterol. Plozasiran cut triglycerides and apolipoprotein C-III levels, lowering pancreatitis risk. A 14-year study linked smoking among youth to cardiac abnormalities. Baseline hsCRP, LDL-C, and Lp(a) were strong predictors of 30-year outcomes in women. Alternative LDL-lowering strategies matched high-intensity statins in effectiveness of LDL-C lowering and reduced diabetes risk. Early combination lipid lowering therapy improved outcomes post-myocardial infarction. Nordic and Mediterranean diets were linked to lower atherosclerotic CVD risk. The findings from the 2024 ESC Congress highlight significant advancements in CVD prevention, including novel lipid-lowering therapies, biomarker-based risk prediction, and lifestyle interventions. These studies underscore the importance of early and personalized treatment strategies to mitigate long-term cardiovascular risk.